Prescripto AI 2.0 — Executive Summary Dossier

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Live Deployed Backend: `https://prescripto-2-0.onrender.com`
Canonical Spec Reference: docs/PROJECT-SPEC.md, docs/Architecture.md
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1. System Identity and Vision

Prescripto AI 2.0 is an enterprise-grade medical document intelligence and medication-safety decision-support system. Designed to address the clinical risks caused by illegible handwritten medical prescriptions, Prescripto converts scanned prescriptions into structured clinical records, maps each drug to canonical pharmaceutical identities, evaluates drug safety boundaries (duplicate therapies, dosage thresholds, and patient allergy contraindications), and presents a provenance-backed workspace to qualified clinicians for mandatory sign-off.

Core Engineering Axiom

> ***"Upstream uncertainty must never silently become downstream clinical certainty."***

In traditional healthcare software, optical character recognition (OCR) and machine learning engines frequently make statistical guesses when encountering faded ink or cursive handwriting. If an OCR model is 40% sure that a word is "Metformin", conventional systems round up and treat it as factual, generating a dangerous illusion of certainty. Prescripto AI 2.0 structurally rejects this paradigm: every extracted field preserves its calibrated confidence score and propagates through an explicit 4-state uncertainty taxonomy (`CLEAR` , `AMBIGUOUS` , `UNREADABLE` , `NOT_PRESENT`). Ambiguity is preserved all the way to the clinician interface.

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2. Architectural Blueprint & Operational Anatomy

The system is constructed as a **Hexagonal Clean Architecture (Modular Monolith)** with an asynchronous distributed queue:

```
graph TD
    User([Patient / Clinician / Operator]) -->|HTTPS / REST| Web[React 19 + Vite + Tailwind Fror
    Web -->|VITE_API_BASE_URL| API[FastAPI Modular Monolith]

    subgraph Core Persistence & Storage
        API --> DB[(PostgreSQL 16 Engine)]
        API --> S3[(MinIO / AWS S3 Object Store)]
    end

    subgraph Distributed Processing Engine
        DB -->|FOR UPDATE SKIP LOCKED| Worker[Fenced Asynchronous Worker Engine]
        Worker --> P1[Stage 1: Ingestion & SHA-256 Check]
        P1 --> P2[Stage 2: Image Quality & Blur Gate]
        P2 --> P3[Stage 3: Vision Recognition Gemini / RapidOCR]
        P3 --> P4[Stage 4: Line Extraction & Token Mapping]
        P4 --> P5[Stage 5: 8-Field Clinical Structure Parser]
        P5 --> P6[Stage 6: Trigram Fuzzy Normalization CDSCO]
        P6 --> P7[Stage 7: Clinical Safety Screening]
        P7 --> P8[Stage 8: Report Assembly & Trigger Routing]
    end

    subgraph Clinician Oversight & Review
        P8 -->|Status: REQUIRES_REVIEW| ReviewQueue[Clinician Verification Workspace]
        ReviewQueue -->|Approve / Override / Escalate| Doctor([Qualified Medical Officer])
    end
```

Key Engineering Pillars

1. **Asynchronous Non-Blocking Ingestion**: Ingestion returns HTTP `202 Accepted` in sub-100ms with a deterministic `analysis_id` and prescription tracking token.
2. **PostgreSQL-Backed Queue with Monotonic Generation Fencing**: Prevents split-brain worker collisions and duplicate executions using `SELECT ... FOR UPDATE SKIP LOCKED` and lease generation tokens.
3. **Multi-Model Vision & RapidOCR Fallback**: Primary perception runs via Gemini 3.5 Flash (`gemini-3.5-flash`), backed by a secondary fallback model (`gemini-3.1-flash-lite`), and an air-gapped on-premise local fallback using ONNX PaddleOCR (`RapidOCRRuntime`).
4. **Deterministic Clinical Screening Rules**: Safety logic is separated entirely from probabilistic models. Drug duplicates, active ingredient overlap, dosage limits, and patient allergen contraindications run through audited deterministic domain rules.
5. **Zero-PHI Structured Logging**: Operational logs pass through a strict field-level whitelist filter (`structlog`) dropping any patient health identifiers before emission.
6. **DPDPA 2023 Verifiable Deletion Protocol**: A 4-phase cryptographic deletion engine guarantees verifiable purging of prescription scans and database records with signed HMAC manifests stored in a retention vault.

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3. High-Level Technology Inventory

Layer	Primary Technology	Exact Version	Deployment Target	Role / Purpose
Frontend UI	React, TypeScript, Vite, TailwindCSS	React `^19.3.0`, Vite `^8.3.0`	Vercel (SPA)	High-contrast, responsive clinician and patient portal
Backend API	Python, FastAPI, Starlette	Python `3.11`, FastAPI `^0.115.0`	Render Web Service	RESTful API endpoints, CORS handling, JWT auth
Relational DB	PostgreSQL	`16-alpine`	Render Managed Postgres	Relational data integrity, CHECK constraints, job queue
ORM & Migrations	SQLAlchemy, Alembic	SQLAlchemy `2.0.35`, Alembic `1.20.0`	Backend / Docker	Schema management, connection pooling, migrations
Object Storage	MinIO (local dev) / AWS S3	`boto3 ^1.35.0`	MinIO / S3 Bucket	Prescription image vault with HMAC presigned URLs
Vision & ML	Google GenAI SDK, RapidOCR	`google-genai ^2.14.0`, RapidOCR	Docker Worker / Cloud	Dual-tier handwriting transcription & entity tagging
Authentication	PyJWT, bcrypt, Cryptography	`PyJWT ^2.9.0`, `bcrypt ^4.2.0`	Backend Auth Module	RS256 asymmetric JWT signing and password hashing
Voice / Multilingual	Web Speech API, React Context	Native Browser API	Client Browser	Speech synthesis in English, Hindi, Gujarati, Marathi

4. Empirical Evaluation and Benchmark Summary

- **Automated Backend Test Suite**: 197 automated test cases spanning unit, integration, and contract tests.
- **Database Concurrency Benchmark**: Verified under simulated multi-worker contention using PostgreSQL row-level locks.
- **Formulary Master Database**: Seeded with 107 verified Indian and international medications spanning antibiotics, analgesics, antacids, antihistamines, cardiac, diabetes, respiratory, and thyroid formulations.
- **Adversarial Image Robustness**: Benchmarked against blur (Laplacian variance < 100.0), extreme underexposure, rotated scans (90°/180°), and corrupted payloads.

5. Scope & Explicit Clinical Safety Boundaries

Supported Capabilities (Implemented)	Explicitly Out of Scope / Excluded by Design
✅ Prescription image ingestion, hashing, and storage	❌ Autonomous medical diagnosis or disease staging
✅ 8-field structured extraction with confidence states	❌ Autonomous drug prescribing or altering treatments
✅ Exact canonical drug matching against CDSCO master	❌ Replacement for certified pharmacy dispensing systems
✅ Duplicate drug and active ingredient detection	❌ Pharmacokinetic / Pharmacodynamic blood concentration modeling
✅ Single-dose threshold safety evaluation	❌ Unverified external web-scraping for drug evidence
✅ Patient allergen cross-referencing against clinical history	❌ Clearing software as an FDA/CDSCO regulated Medical Device
✅ Human-in-the-loop review queue and clinical sign-off	❌ Unsupervised clinical decision making